

Algebra 1: 2nd Semester Final Review

Chapter 8 & 9

State the transformations that occurred from $p(x) = x^2$ to the given function.

1. $g(x) = -2(x - 4)^2 + 5$

Reflection
Vertical Stretch
Right 4
UP 5

2. $f(x) = (x + 3)^2 - 2$

Left 3
Down 2

Find the zeros of the function.

3. $f(x) = 3x^2 - 6x - 45$

$3(x^2 - 2x - 15)$
 $3(x - 5)(x + 3)$

$x = \{-3, 5\}$

4. $k(x) = 2(x - 1)(x + 7)$

$x = 1$ $x = -7$

$x = \{-7, 1\}$

5. $g(x) = x^3 - 16x$

$x(x^2 - 16)$
 $x(x - 4)(x + 4)$

$x = \{-4, 0, 4\}$

Solve the equation:

6. $r(r - 10) = 0$

$r = 0$

$r = 10$

$r = \{0, 10\}$

7. $(h - 8)^2 = 0$

$h = 8$

$h = \{8\}$

8. $-2v(v + 1) = 0$

$-2v = 0$ $v + 1 = 0$

$v = 0$ $v = -1$

$v = \{-1, 0\}$

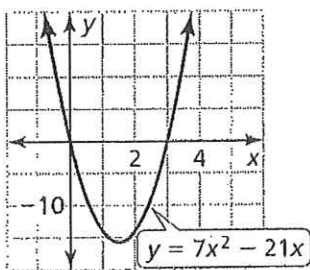
9. $(x - 2)(3x - 1) = 0$

$x = 2$ $x = \frac{1}{3}$

$x = \{\frac{1}{3}, 2\}$

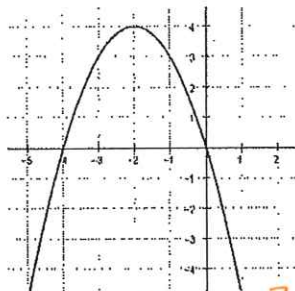
10. Find the solutions of each function.

a)



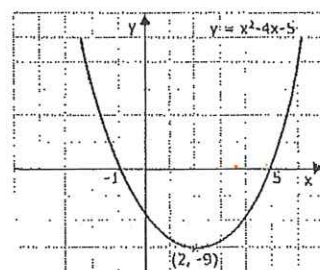
$x = \{0, 3\}$

b)



$x = \{-4, 0\}$

c)



$x = \{-1, 5\}$

Find the vertex and axis of symmetry and vertex of the graph of the function.

12. $f(x) = 2x^2 - 20x + 51$

$x = \frac{20}{2(2)} = \frac{20}{4} = 5$

$2(5)^2 - 20(5) + 51$

Vertex: $(5, 1)$

A.O.S.: $x = 5$

13. $f(x) = -\frac{1}{2}(x - 3)^2$

Vertex: $(3, 0)$

A.O.S.: $x = 3$

$$y = a(x-h)^2 + k$$

11. Consider the graph of the function f . Find the following:

Domain: $\mathbb{R}$

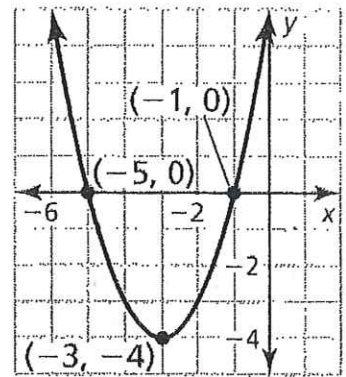
Range: $y \geq -4$

Zeros: $x = \{-5, -1\}$

Vertex: $(-3, -4)$

AS: $x = -3$

Write the function in vertex form: $y = (x+3)^2 - 4$



Solve the equation using any method. Leave answer in simplified radical form.

16. $(2x+1)^2 = 36$

$$\begin{aligned} (2x+1)(2x+1) &= 36 \\ 4x^2 + 2x + 2x + 1 &= 36 \\ 4x^2 + 4x - 35 &= 0 \\ -\frac{4 \pm \sqrt{4^2 - 4(4)(-35)}}{2(4)} &= \frac{-4 \pm \sqrt{576}}{8} \\ &= \frac{-4 \pm 24}{8} = \left\{ \frac{20}{8}, \frac{-28}{8} \right\} \\ &= \left\{ \frac{5}{2}, -\frac{7}{2} \right\} \end{aligned}$$

18. $x^2 - 3x - 40 = 0$

$$\begin{aligned} \frac{3 \pm \sqrt{(-3)^2 - 4(1)(-40)}}{2(1)} \\ \frac{3 \pm \sqrt{169}}{2} = \frac{3 \pm 13}{2} = \left\{ -5, 8 \right\} \end{aligned}$$

17. $(4x+5)^2 = 9$

$$\begin{aligned} 4x+5 &= \pm 3 \\ 4x+5 &= 3 & 4x+5 &= -3 \\ 4x &= -2 & 4x &= -8 \\ x &= -\frac{1}{2} & x &= -2 \\ x &= \left\{ -\frac{1}{2}, -2 \right\} \end{aligned}$$

19. $-5x^2 = -25$

$$\begin{aligned} \sqrt{x^2} &= \sqrt{5} \\ x &= \left\{ \pm \sqrt{5} \right\} \end{aligned}$$

How many real solutions does each quadratic function have? (Hint: find the discriminant)

20. a. $4x^2 + 4x + 1$

$$\begin{aligned} 4^2 - 4(4)(1) \\ 16 - 16 \\ 0 \end{aligned}$$

1 solution

b. $2x^2 - x - 1 = 0$

$$\begin{aligned} (-1)^2 - 4(2)(-1) \\ 1 + 8 \\ 9 \end{aligned}$$

2 solutions

21. Find the vertex: $k(x) = -(x-3)(x+2)$

$$\begin{aligned} (-x+3)(x+2) \\ -x^2 - 2x + 3x + 6 \\ -x^2 + x + 6 \end{aligned}$$

$$x = \frac{-1}{2(-1)} = \frac{1}{2}$$

$$-(-\frac{1}{2} - 3)(-\frac{1}{2} + 2)$$

$$x = \left(\frac{1}{2}, \frac{25}{4} \right)$$

22. A ball is dropped out a window. The function $h = -16t^2 - 8t + 25$ represents the height (in feet) of the ball after t seconds. What is the initial height of the ball?

$$h = -16(0)^2 - 8(0) + 25$$

$h = 25$ feet